SIM1002 Calculus I
Tutorial 8
Applications of Derivatives

Exercise 4.3

Identifying Extrema

In Exercises 1-3:

(a) Find the open intervals on which the function is increasing and those on which it is decreasing.

(b) Identify the function's local and absolute extreme values, if any, saying where they occur.

1. $h(x) = 2x^3 - 18x$

2. $g(x) = x^2\sqrt{5-x}$

3. $k(x) = x^{\frac{2}{3}}(x^2 - 4)$

In Exercises 4-6:

(a) Identify the function's local extreme values in the given domain, and say where they occur.

(b) Graph the function over the given domain. Which of the extreme values, if any, are absolute?

4. $f(t) = t^3 - 3t^2,\ (-\infty, 3]$

5. $f(x) = \sqrt{x^2 - 2x - 3},\ [3, \infty)$

6. $g(x) = \frac{x^2}{4-x^2},\ (-2,1]$

In Exercises 7 and 8:

(a) Find the local extrema of each function on the given interval, and say where they occur.

(b) Graph the function and its derivative together. Comment on the behavior of f in relation to signs and values of f'.

7. $f(x) = \sin x - \cos x\,,\ 0 \le x \le 2\pi$

8. $f(x) = -2x + \tan x\,, -\frac{\pi}{2} < x < \frac{\pi}{2}$

Exercises 4.4

In Exercises 1-5, graph the function using appropriate methods from the graphing procedures presented. Identifying the coordinates of any local extreme points and inflection points. Then find coordinates of absolute extreme points, if any.

1. $y = \frac{x^2-4}{2x}$

2. $y = \frac{4x}{x^2+4}$

3. $y = \cos x + \sqrt{3} \sin x\,,\ 0 \le x \le 2\pi$

4. $y = x^{\frac{2}{3}}(x-5)$

5. $y = |x^2 - 2x|$

Exercises 4.7

Finding Antiderivatives

In Exercises 1-3, find an antiderivative for each function. Check your answers by differentiation.

1. (a) $\frac{4}{3}x^{\frac{1}{3}}$ (b) $\frac{1}{3}\left(\frac{1}{\sqrt[3]{x}}\right)$ (c) $\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}$

2. (a) $\pi \cos(\pi x)$ (b) $\frac{\pi}{2}\cos\left(\frac{\pi x}{2}\right)$ (c) $\cos\left(\frac{\pi x}{2}\right) + \pi \cos x$

3. (a) $\sec x \tan x$ (b) $4\sec(3x)\tan(3x)$ (c) $\sec\left(\frac{\pi x}{2}\right)\tan\left(\frac{\pi x}{2}\right)$

Finding Indefinite Integrals

In Exercises 4-8, find the most general antiderivative or indefinite integral. Check your answer by differentiation.

4. $\int \left(\frac{\sqrt{x}}{2} + \frac{2}{\sqrt{x}}\right) dx$

5. $\int \frac{(4+\sqrt{t})}{t^3} dt$

6. $\int \left(-\frac{\sec^2 x}{3}\right) dx$

7. $\int x^{\sqrt{2}-1}\, dx$

8. $\int \frac{cosec\ \theta}{cosec\ \theta - \sin\theta}\, d\theta$